Dual Effects of Hybrid Working on Performance: More Work Hours or More Work Time

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Abstract. Work in software development companies has become increasingly hybrid with employees altering days of working in the office with days of working remotely from home. Yet, little is know about the efficiency of such way of working because the current scale of remote working is unprecedented. In this paper, we present our findings from a company-wide survey at Storebrand - a large-scale Norwegian fintech company, focusing on perceived performance. Our analysis of 192 responses shows that most employees report being able to perform the planned tasks. Further, half of respondents perceive to have increased work hours. Through qualitative analysis of open-ended commentaries of respondents we learned that remote working has dual effects on the perceived work hours - some employees report working longer hours and others report having more work time due to efficient use of the time throughout the day. Finally, we recommend managers to discuss and address the concerning habits of employees caused by increased connectivity and inability to stop working, before these lead to burnout and disturbances in the work/life balance.

Keywords: Hybrid · Flexible · Remote · Performance · Work hours

1 Introduction and Related Work

Many software development companies today have become places of hybrid working with polices that institutionalize remote-first culture with little onsite presence, hybrid work culture often with rather flexible onsite presence, and office-first culture with frequent office presence yet some flexibility for working remotely [10]. As a result, individuals can choose, at least to some degree, when to work remotely and when to have office-based work [3,7,10,13].

Hybrid working, that is, alternating between working at the office and working remotely, is said to bring the best of the two worlds: remote and onsite working. The documented benefits of remote working include reduced commute time,

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better conditions for focused work, a more comfortable work environment, and better work-life balance [12,13]. In turn, office working provides the opportunities to socialise face-to-face, facilitate efficient brainstorming and problem-solving sessions, provide help and onboard team members, have spontaneous interactions, and strengthen team cohesion and psychological safety, all the aspects that are reported to be missing in fully remote work rhythm [15]. Yet, the ability to tap into these benefits is conditioned by the level of co-presence. Individual office presence, does not guarantee that people meet and interact, if office days are not synchronized [8]. Even when the majority of team members opt for office-based work, many teams maintain a hybrid structure to accommodate those who prefer arrangements that are more remote in nature [13].

The lack of co-presence significantly affects communication, as well as the effectiveness of the regular coordination mechanisms when used in a computer-mediated fashion [5,14]. When some people are working from home and others are working from the office, conducting spontaneous discussions is difficult and sub-groups emerge within the team [15]. Hybrid teams are found to have an increased need for coordination which shifted from informal, spontaneous face-to-face coordination towards more formal, consistent, technology-mediated synchronous (i.e. meetings) and asynchronous (e.g. shared documents; Slack messages) communication [14]. Further, agile ways of working and rituals that heavily rely on co-presence intend to facilitate collaboration and enable individual performance. Early studies of hybrid agile teams show that many agile practices held remotely are compromised [14]. The effects of hybrid work on performance is thus to be understood.

The key questions for leaders today center around the ways to find a balance between individual, team, and organizational needs. In particular, many companies try to understand the value and benefits of office presence vs remote working, and how much office presence or co-presence in work groups is sufficient to maintain performance, and keep employees satisfied.

Motivated by the above-mentioned challenges we seek to answer the following research question: ***How is perceived performance affected by hybrid working?*** To address our research question, we conducted a case study in a large-scale Norwegian fintech company.

2 Research Methodology

Empirical Background: In this workshop paper, we report our findings based on the data collected at Storebrand, a Norwegian financial services company that offers pension, savings, insurance, and banking products to both the private and the business markets. Based on a strategic work launched at the start of the COVID-19 pandemic, Storebrand committed to changing the corporate work policy to explicitly permit flexible working. Storebrand had used agile methods for over a decade and can be understood as a mature agile organization.

Hybrid strategy at Storebrand is based on a belief that a one-size-fits-all solution does not exist and even one lasting solution for the same team can

be utopic. Therefore, the company intentionally fosters team-based discussions over rules to build lasting habits around sound reasoning and cooperation rather than strict policies. Despite a very flexible policy, Storebrand management does believe in the importance of an attractive office and office-based working [9]. However, instead of asking how to attract people to the office, Storebrand defines three core questions to guide their hybrid strategy:

- How can we organize our work as efficiently as possible to create the products and services that our customers need?
- How can we create the most engaging and inclusive workplace possible?
- How can we ensure that all employees experience mastery and development?

Data Collection and Analysis: We collected the data through an online survey. The survey was open during June 12–23, 2023, for all 2,166 employees and gathered 1072 responses (over 50% response rate). Here, we report the analysis of a subset of data from the software engineering department at Storebrand.

The software development department employed 399 software engineers and business people who are involved in developing digital financial products and work according to agile methods - large-scale Scrum and ScrumBan. Employees are situated both in Norway and Sweden. Our sample includes 192 responses (48% response rate). We focus on questions related to individual work rhythm, perceived task performance, and perceived changes in work hours. The responses were collected on a 5-point scale, including the response alternatives on a range between "Not at all" to "Completely", "Very dissatisfied" to "Very satisfied", "To a very small extent" to "To a very large extent". The subset of survey questions relevant for this paper is included in the Appendix.

Stratified analysis was performed for varying degrees of office presence. Responses to open-ended questions were analyzed qualitatively using thematic coding and generated themes that help explain the quantitative results.

3 Results

In the following, we present the results from surveying employees in the software engineering department of Storebrand about their perceived performance. In Fig. 1, we start with the details about the survey respondents and their demographic information, followed by their hybrid work rhythm in Fig. 2. We then, in Fig. 3, present the distribution of responses to the performance question. This is, to the employees' ability to complete their tasks, stratified by the hybrid work rhythm (proportion of the office presence vs remote working). We then seek explanation for our results through the combined analysis of both quantitative and qualitative responses to related questions.

Our survey represents experiences from predominantly Norwegian employees from the software engineering department at Storebrand. 1/3 of the respondents are female, and 2/3 are male respondents. The sample has a good representation of employees in different age groups. With respect to commute time, the majority

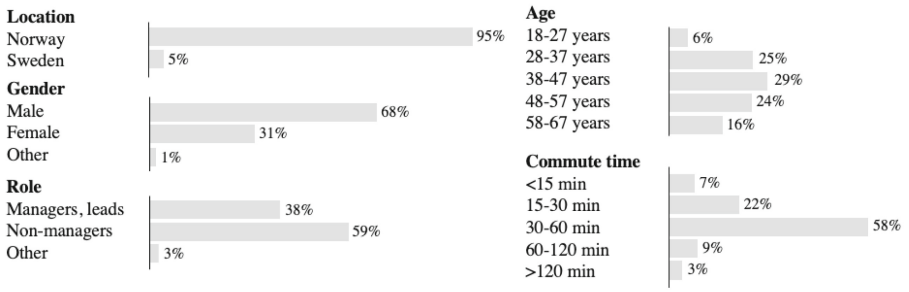


Fig. 1. Overview of the respondents

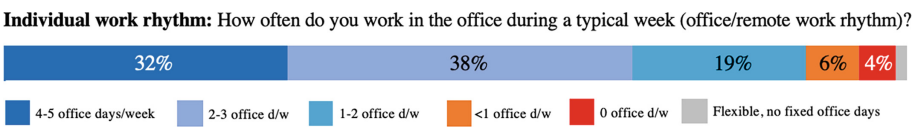


Fig. 2. Hybrid work rhythms

of respondents need 30–60 min to reach the office and roughly 1/3 lives within 30 min commute distance.

One of the interesting findings in the survey is related to the hybrid work rhythms (See Fig. 2). Despite Storebrand's flexible work policy, we found that 70% of respondents are working in the office half of the time or more. Only 10% do not visit the office regularly every week, four of which are working fully remotely.

In Fig. 3, we illustrate the respondents' ability to complete planned tasks. We see that the majority of respondents are able to accomplish their tasks to a large or very large extent. This means that hybrid working does not hinder performance. Although employees with fully flexible and fully remote work rhythms report higher performance scores, this result shall be taken with a grain of salt because there are just a few respondents in these categories. The scores in the categories with higher number of respondents are quite similar, with slightly better scores in the category of employees having one to two days per week working in the office.

To better understand the variance in perceived performance, we asked the respondents about the changes in their work hours due to hybrid work, which directly affect the ability to accomplish the planned work tasks. Further, we explored the free-text responses related to performance questions and identified further impact factors inherent in the hybrid ways of working.

The survey results suggest that half of respondents (50%) do not perceive any change in the work hours and a marginal share of respondents feel that the number of hours has slightly decreased (1%), as shown in Fig. 4. Yet, the other half of respondents (49%) report an increase in work hours (29% responded that work hours Slightly increased and 20% that work hours Increased, see Fig. 4).

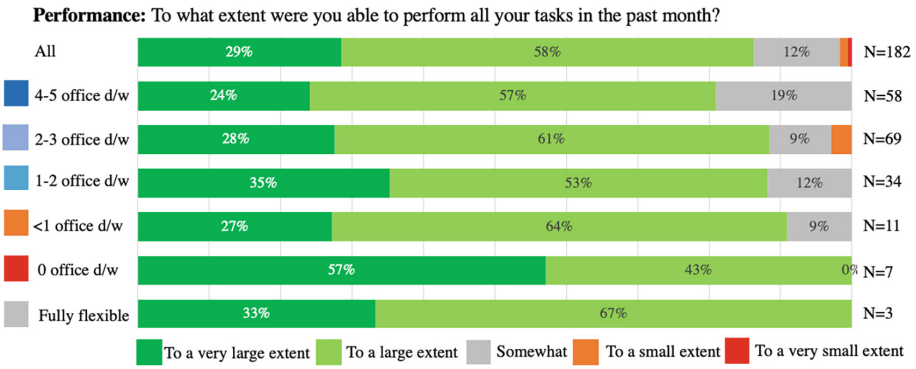


Fig. 3. Ability to complete tasks

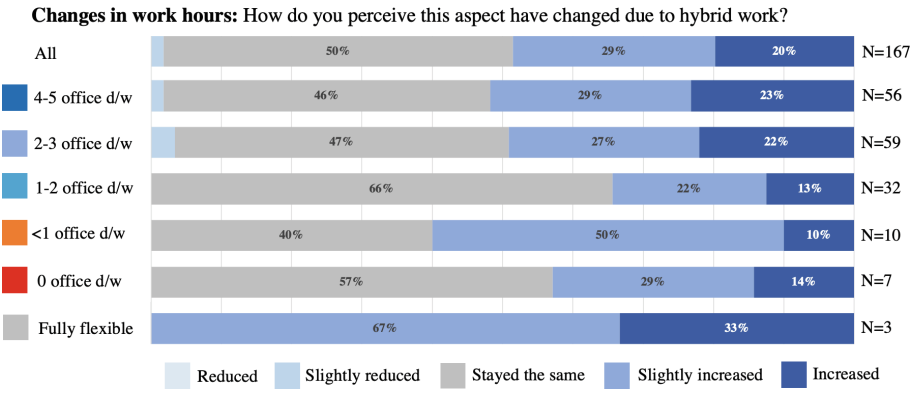


Fig. 4. Changes in work hours and factors explaining these changes

This can also explain our findings with respect to positive performance scores as more can be done when working longer hours. Detailed analysis shows that the lowest number of respondents with increased work hours appear in the category working 1–2 days in the office. Additional commentaries regarding the work hours (from 26 respondents) pointed towards the different aspects of working remotely (See Fig. 4). Among these, we found, for example, positive effects, such as the lack of commute time that led to having more work time. However, the most frequent reason contributing to having more or longer work hours was increased connectivity, which related to higher mobilisation and lower threshold for contacting the colleagues but also a more blurry boundaries between work and private life. Descriptions in this category were sometimes alarming, indicating that people sacrifice their free time and privacy for performing the work. Yet, a fully onsite work is not an escape from increased work hours either. In the category reporting 4–5 days weekly office presence, 52% report having increased work hours.

4 Concluding Discussion

In this paper, we presented our findings from studying hybrid work rhythms and individual performance. We found that a company with flexible work policy had, what one could judge as an unexpectedly high office presence. We also found that the alternation of office presence and remote work results in most employees (87%) reporting being able to perform their tasks to a large or very large extent. This is in line with many other studies that report positive impacts of remote working on productivity due to schedule flexibility, freedom from interruptions, and time saved on commuting [4,7,12,13,16]. Some studies that try to understand why higher productivity occurs, point out that the increased outputs can be attributed to employees working longer hours, suggesting that productivity increases might be illusive [1,2,13,14]. In our case, we found that high performance can be indeed explained by the changes in the work hours - 52% of respondents report an increase in work hours. Similarly, de Souza Santos et al. found in their survey study that half of respondents acknowledged working overtime, with some (3%) even exceeding five extra hours per week [13]. The reasons for increased work hours in our study echo the findings by de Souza Santos et al.. We found increased work hours to be associated with a more efficient use of time (contributing to having more work time), but also due to the negative impacts of increased connectivity and not having a strict work/life separation (contributing to longer work hours). Along with concerns about the tendency to work longer hours, some researchers warn about the threats to work/life balance [6,11,13].

At the same time, our results show that an increase in work hours is not caused by increased remote working, as even respondents with the highest office presence reported negative changes in work hours. Further, we discovered that the lowest increase in work hours was reported by those only in the office 1–2 days a week. This may be due to having more control over work schedule, and fewer interruptions when performing individual tasks. Since there is no clear link between work arrangement and longer hours, we conclude that these concern job pressure and individual habits related to increased connectivity. One recommendation to address the emergent habits for those working from home is to implement a "hard stop" practice-a preset time for logging off or turning off the computer-which we found effective in our earlier study [11]. For those working in the office, the availability of focused work places might be a solution.

In our future work, we will continue monitoring the changes in the work hours and focus on understanding the work/life balance. While this study reports findings on an individual level, further work shall also focus on understanding the impact of remote and hybrid working on teams.

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A Appendix: Survey Questions

In the following, we provide the subset of survey questions used in the analysis.

Location: Which country do you primarily work from? (Norway/Sweden/Other countries)

Gender: Your gender (Female/Male/Other/Prefer not to disclose)

Role: Do you have responsibility for other's work? (Yes, I have personnel responsibility for 1+ persons/Yes, I have managerial responsibility without personnel responsibility/No/Not sure)

Age: How old are you? (18–27 years/28–37 years/38–47 years/48–57 years/58–67 years/Older than 67 years)

Commute: How much time do you spend commuting to the office (one way)? (Less than 15 min/Between 15 and 30 min/Between 30 min and 1 h/Between 1 and 2 h/More than 2 h/Not relevant)

Work rhythm: How often do you work in the office during a typical week? (I never or almost never work in the office/Less than one day a week/1–2 days a week/2–3 days a week/4–5 days a week/I don't have fixed days I work in the office)

Performance: To what extent were you able to perform all your tasks in the past month? (To a very small extent/To a small extent/Somewhat/To a large extent/To a very large extent)

Changes in work hours: How do you perceive these aspects have changed due to hybrid work - Number of hours you work? (Have been reduced/Slightly reduced/Remained the same/Slightly increased/Have increased/Not relevant)

Explanation: If you experienced any changes, what is the reason for it?

References

1. Bloom, N., Liang, J., Roberts, J., Ying, Z.J.: Does working from home work? Evidence from a Chinese experiment. Q. J. Econ. **130**(1), 165–218 (2015)
2. Chesley, N.: Technology use and employee assessments of work effectiveness, workload, and pace of life. Inf. Commun. Soc. **13**(4), 485–514 (2010)
3. Conboy, K., Moe, N.B., Stray, V., Gundelsby, J.H.: The future of hybrid software development: challenging current assumptions. IEEE Softw. **40**(02), 26–33 (2023)

4. Conradie, W.J., De Klerk, J.J.: To flex or not to flex? Flexible work arrangements amongst software developers in an emerging economy. SA J. Hum. Resour. Manag. **17**(1), 1–12 (2019)
5. Espinosa, J.A., Carmel, E.: The impact of time separation on coordination in global software teams: a conceptual foundation. Softw. Process: Improve. Pract. **8**(4), 249–266 (2003)
6. Gorjifard, R., Crawford, J.: Working from home: impact on wellbeing and work-life balance. N. Z. J. Employ. Relat. **46**(2), 64–78 (2021)
7. Malhotra, A.: The postpandemic future of work (2021)
8. Moe, N.B., Ulsaker, S., Hildrum, J.M., Smite, D., Ay, F.C.: Understanding the difference between office presence and co-presence in team member interactions. In: Proceeding of the 57th Hawaii International Conference on System Sciences (HICSS 2024). AIS Electronic Library (2024)
9. Smite, D., Klotins, E., Moe, N.B.: What attracts employees to work onsite in times of increased remote working? IEEE Softw. 1–5 (2024). https://doi.org/10.1109/MS.2024.3375964
10. Smite, D., Moe, N.B., Hildrum, J., Gonzalez-Huerta, J., Mendez, D.: Work-from-home is here to stay: call for flexibility in post-pandemic work policies. J. Syst. Softw. **195**, 111552 (2023)
11. Smite, D., Moe, N.B., Klotins, E., Gonzalez-Huerta, J.: From forced working-from-home to voluntary working-from-anywhere: two revolutions in telework. J. Syst. Softw. **195**, 111509 (2023)
12. Smite, D., Tkalich, A., Moe, N.B., Papatheocharous, E., Klotins, E., Buvik, M.P.: Changes in perceived productivity of software engineers during COVID-19 pandemic: the voice of evidence. J. Syst. Softw. **186**, 111197 (2022)
13. de Souza Santos, R., Grillo, W., Cabral, D., de Castro, C., Albuquerque, N., França, C.: Post-pandemic hybrid work in software companies: findings from an industrial case study. arXiv e-prints, pp. arXiv–2401 (2024)
14. de Souza Santos, R.E., Ralph, P.: A grounded theory of coordination in remote-first and hybrid software teams. In: Proceedings of the 44th International Conference on Software Engineering, pp. 25–35 (2022)
15. Tkalich, A., Šmite, D., Andersen, N.H., Moe, N.B.: What happens to psychological safety when going remote? IEEE Softw. (2022)
16. Vargas Llave, O., et al.: The rise in telework: impact on working conditions and regulations (2022)

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